

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 16

Let (X, Y) be a random vector.

Def: $\text{Var}(X|Y) = h(Y)$ where $h(y) = E((X - E(X|Y = y))^2 | Y = y)$
 $= E(X^2 | Y = y) - (E(X | Y = y))^2$.

Theorem: $\text{Var}(X) = E(\text{Var}(X|Y)) + \text{Var}(E(X|Y))$.

Example 1: Let $X_0, X_1, X_2, \dots, X_n$ be a sequence of i.i.d. RVs with mean μ and variance σ^2 . Let $N \sim \text{Bin}(n, p)$, independent of $\{X_i\}$.

Define $S = \sum_{i=0}^N X_i$. Find $\text{Var}(S)$.

Computing Probability by Conditioning

$$P(E) = \begin{cases} \sum_y P(E|Y=y)P(Y=y) & \text{for } Y \text{ discrete} \\ \int_{-\infty}^{\infty} P(E|Y=y)f_Y(y)dy & \text{for } Y \text{ continuous.} \end{cases}$$

Example 2: Let X and Y be independent CRVs having PDFs f_X and f_Y , respectively. Compute $P(X < Y)$.

Example 3: Let X and Y be i.i.d. CRVs having common PDF f . Then $P(X < Y) = P(X > Y) = 0.5$. And $P(X = Y) = 0$.

Example 4: Suppose X and Y are two independent RVs, either discrete or continuous. What is the distribution of $X + Y$?

Def: Let (X, Y) be a random vector. Then

$$E(h(X, Y) | (X, Y) \in A) = \frac{E(h(X, Y)I_A(X, Y))}{P((X, Y) \in A)}.$$

Example 5: $X \sim \text{Exp}(1)$. Find $E(X | X \geq 2)$.

Example 6: (X, Y) is uniform on unit square. Find $E(X | X + Y > 1)$.